

## Econometrics in R

### Day 1

Opensource-see code

National family health service-

MDPI-niti aayog

Economic survey

**socio economic high resolution data**

Development data lab

### Linear regression

Variance-

Covariance-

Random variable - chance experiment - coin toss- clear outcomes - numeric value outcome of a random experiment

Discrete and Continuous random variables

$$\frac{1}{2} \times 1 + \frac{1}{2} \times -1$$

$$= \text{mean} = 0.2$$

Law of large numbers- as the sample size increases the mean get closer to expected value (population mean)

Chances of winning/losing become equal

$$\frac{1}{2} \rightarrow 0.5$$

$$\frac{1}{2} \rightarrow 0.5$$

$$0.5 \times 1 + 0.5 \times -1$$

Relative frequency converges as the sample size increases

**Expected value of a random variable** - limit to which the sample tends as the sample gets bigger and bigger

Average of an outcome is the expected value

Sample mean does not equal to population mean for all samples - sample mean is a random variable

As the sample size increases it moves towards pop mean- consistency

Quality of sample > quantity

Representative sample- every element of the sample has the same chance of being selected

Random sample- every element has an equal chance of being included in the sample → larger sample

Sampling design- purposive sampling and snowball sampling

Outcomes with different sampling designs cannot be compared

| X  | P(x) |                                       |
|----|------|---------------------------------------|
| 1  | 0.5  | $0.5 \times 1 + 0.5 \times -1$        |
| -1 | 0.5  | $P(x=1) \times 1 + P(x=-1) \times -1$ |

Eg. roll a die

Limit of relative frequency is  $\frac{1}{6}$

| X        | P(x) |
|----------|------|
| 1        | 1/6  |
| 2        | 1/6  |
| 3        | 1/6  |
| 4        | 1/6  |
| 5        | 1/6  |
| 6        | 1/6  |
| E(x)=3.5 |      |

Variance- average distance of the observation from the mean - always +ve -covariance of x with itself

Put a root- standard deviation

As sample size increases, sample variance will get closer to population variance

Degrees of freedom- set of data points- find mean - no of data points that have independent values is df

Using data points for estimation- no of independent data points

Data points are dependent on mean

Every additional point of data is more info- sample estimate gets better

R

Imagine a lot of toss - samples of size pi

```
library(ggplot2)
library(dplyr)
population<-rep(c(1,-1),times=10000)
x<-sample(population,5,replace=FALSE)
space<-c(rep(0.10000)
i=1
repeat {
x<-sample9population,5,replace=FALSE)
space[i]=mean(x)
i=i+1
if(i>10000)
break
}
space<-data.frame(space)
p<-ggplot(space,aes(x=space))=geom_density(aes(x=space)fill="green
```

Population size -10000

Sample size - 5

Repeat the same exercise - get 10000 sample means and plot them - get bell shaped curve

Get 10000 0 slots - p of sample mean

Sample mean is same as pop mean

Unbiased estimator of the pop mean - sample

Overestimation and underestimation in single sample but overall the mean is 0

Increase sample to 500-as sample increased pop mean becomes more accurate

Compare with previous

```
sample1<-c(rep(0,10000))
```

```
i=1
```

```
repeat {
```

```
x<-sample(population,500,replace=FALSE)
```

Covariance-

|   | M     | L | M-M- | L-L- | (M-M-)(L-L-) |
|---|-------|---|------|------|--------------|
| 1 | 7     | 4 | 1    | -2   | -2           |
| 2 | 5     | 8 | -1   | 2    | -2           |
| 3 | 8     | 8 | 2    | 2    | 4            |
| 4 | 5     | 7 | -1   | 1    | -1           |
| 5 | 5     | 3 | -1   | -3   | 3            |
|   | ----- |   |      |      |              |
|   | 6     | 6 |      |      |              |

$$2/4 = 0.5$$

Individual —heterogeneity

$$Y_i = \alpha + \beta X_i + e$$

Find covariance

$$\text{cov}(Y_i, X_i) = \text{cov}(X_i, (\alpha + \beta)) + \text{cov}(X_i, e)$$

$$\begin{aligned}\text{cov}(X, Y_i) &= \text{cov}(\beta X, X) \\ &= \beta \text{cov}(x, x)\end{aligned}$$

$$\begin{aligned}\text{cov}(X_i, Y) &= \beta \text{var}(X) \\ \beta &= \text{cov}(X_i, Y) / \text{var}(x)\end{aligned}$$

Derive  $\hat{\beta}$

## Day 2

Omitted variable bias- bias with the reg variable where your reg should have more variables that are not correlated.

(1000 random numbers, 0 mean, 1 std dev)

From pop 1000 we take sample of 30  
Regression y on x 1000 times  
Get 1000 values of beta hat  
Check distribution  
Type data sampled in console  
I-type 2 error → power of a test  
NAtioal Income (Y,C) - endogeneity

In R — `install.packages("dplyr")`

In R

```
data<-read.csv("class.csv")
attach(data)
summary(data)
cov(a,c)
cov(data)
```

```
data<-read.csv("class.csv")
attach(data)
summary(data)
cov(a,c)
cov(dainstata)
```

```
rm(c)
data<-read.csv("table.csv")
attach(data)
summary(data)
time<-seq(1950,2020,1)
plot(gnpreal~time,type="l")
length(time)
```

```
rm(c)
data<-read.csv("tab.csv")
attach(data)
summary(data)
time<-seq(1950,2022,1)
plot(gnpreal~time,type="l")
```

```
library(dplyr)
library(ggplot2)
data<-read.csv("india.csv")
attach(data)
bankdistance<-ifelse(distance_of_banks==5,1,0)
```

```
sum(bankdistance)
data1<-data.frame(data,bankdistance)
state<-group_by(data1,state_name,district_name)%>%summarize(nobank=mean(bankdistance
))
write.csv(state,"state.csv")
```